Development of a 2-DoF Antenna Radiation Measurement System

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Abstract. Understanding antenna radiation patterns is crucial for optimizing wireless communication systems. Although simulation tools predict radiation patterns during development, environmental conditions in real-world settings substantially influence outcomes. Commercial measurement systems, though precise, are often prohibitively expensive for education. This work presents a low-cost, two-axis computerised system for measuring antenna radiation patterns at 868 MHz, integrating Arduino-based control, stepper motors, and power sensors. Tested with a Yagi-Uda antenna, it yields results consistent with simulations, serving as a practical tool for telecommunications classes. By enabling real-time visualization, this mechatronic platform enhances student understanding of antenna behaviour. Designs will be shared openly to support replication.

Keywords: Antenna Pattern Analysis, Wireless Systems, Mechatronic Education, Telecommunications Training

1 Introduction and System Overview

Antenna radiation patterns are essential in wireless communication engineering, defining signal coverage, interference, and system performance[1]. Simulation software predicts these patterns, yet real-world conditions, such as environmental interference and manufacturing variations, often yield different results[2]. Accurate measurements are thus critical for validating designs. Commercial measurement systems, while precise, are prohibitively expensive and complex for educational settings, limiting students to theoretical studies and simulations [3]. This restricts hands-on learning, making electromagnetic concepts harder to grasp and reducing practical problem-solving opportunities.

To address this challenge, as shown in Figure 1, this work introduces a cost-effective, automated system for measuring antenna radiation patterns. The mechatronic platform uses Arduino-driven NEMA 17 stepper motors(1.8° resolution) and an RF power detector(based on the AD8313 integrated circuit)

for dual-axis antenna rotation, with MATLAB[3] visualisation. Designed in SolidWorks[4], for telecommunications education, it enables students to analyse real-world patterns, fostering practical skills. This approach bridges theoretical knowledge with experimental practice, enhancing classroom learning.



(a) CAD rendering (b) Physical prototype

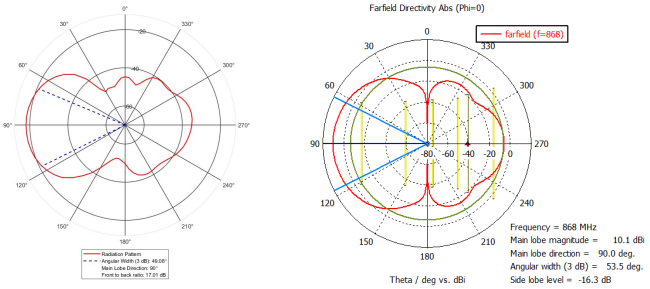
Fig. 1: System for antenna pattern measurement: SolidWorks (left) and assembled (right).

2 Results and Conclusion

The system was evaluated using a Yagi-Uda antenna at 868 MHz, with its measurements compared to CST Studio Suite[5] simulations. Figure 2 illustrates the system's ability to capture radiation patterns, closely aligning with simulated outcomes. The primary lobe was detected at 90°, with a 3 dB beamwidth of 49.08°, near the simulated 53.5°, and front-to-back ratio reduced by 17.01 dB, compared to 16.3 dB theoretically. The noise limits of the RF detector were considered.

Small variations, due to real-world conditions like reflections and stray signals, appeared between measured and simulated patterns, underscoring difficulties in non-ideal test settings. Though external noise impacted precision, the

[3] A campus version of this software is licensed to the Polytechnic Institute of Bragança.
[4] Educational version licensed to the Polytechnic Institute of Bragança.
[5] The free educational version of this software was used in the simulations.



(a) Measured data (b) Simulated outcome

Fig. 2: Measured and simulated radiation patterns.

system consistently captured radiation patterns, proving its value for classroom learning, even with its limitation. This reliability supports its role as an effective tool for teaching antenna concepts in educational environments.

Planned enhancements include filters to block unwanted signals, a balun to stabilize transmission, and optimized impedance alignment. Digital tutorials will be created to spark student interest in antenna principles through interactive visualizations. Sharing system designs openly will encourage collaborative improvements, strengthening its utility as an affordable platform for hands-on telecommunications education and fostering innovation.

Acknowledgment

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Efeito Pelicular e Soluções para Medições de Antenas a 868 MHz

Abstract

Este trabalho aborda o problema das correntes de modo comum em medições de antenas a 868 MHz, causadas pelo efeito pelicular em cabos coaxiais, que induzem irradiação eletromagnética indesejada. Como solução, propõe-se o uso de um balun choke de ar de fabricação própria, permitindo flexibilidade e otimização específica para a faixa de operação. Ressalta-se que a antena utilizada é confeccionada em latão, enquanto o cabo coaxial RG174 possui condutor central de cobre nu e trança externa de cobre estanhado, respeitando as propriedades elétricas de cada material nas análises do efeito pelicular. O dimensionamento do balun é validado por cálculos e medições experimentais, com atenuação de -47 dB no modo comum e perda de inserção de -0,5 dB no modo diferencial. Um capítulo futuro discutirá o casamento de impedâncias para otimizar a transferência de potência.

1 Introdução

As medições do padrão de radiação de antenas em altas frequências, como 868 MHz, requerem cuidados para evitar interferências causadas pelo sistema de alimentação. O uso de cabos coaxiais, como o RG174 (composto por condutor central de cobre nu e trança externa de cobre estanhado), para alimentar antenas teoricamente simétricas, pode introduzir correntes de modo comum na superfície externa da malha do cabo. Tais correntes são agravadas pelo efeito pelicular (*skin effect*), cuja análise neste trabalho leva em consideração tanto a resistividade do cobre (para o cabo) quanto do latão (para elementos da antena, quando relevante). O resultado é a possível irradiação indesejada pelo sistema de alimentação, capaz de distorcer o padrão de radiação obtido em medições e afetar a precisão dos resultados experimentais ([1]; [2]).

Como a linha coaxial é, por natureza, desbalanceada (corrente fluindo majoritariamente no condutor central e na parte interna da malha) e parte das antenas possui características simétricas ou exige alimentação balanceada, torna-se necessária a utilização de um balun choke de ar para bloquear correntes de modo comum e garantir que apenas a corrente diferencial circule no sistema. Com isso, busca-se preservar a fidelidade das medições do padrão de radiação e evitar artefatos decorrentes de irradiação pelo cabo.

Apesar da disponibilidade de baluns comerciais, optou-se pela confecção própria de um balun choke de ar, o que permite ajuste fino das características construtivas para a frequência de operação, flexibilidade em experimentos e aprofundamento didático dos efeitos físicos envolvidos. O casamento de impedâncias, essencial para maximizar a transferência de potência e minimizar reflexões no sistema, será discutido em seção futura deste trabalho. Este trabalho detalha o impacto do efeito pelicular, fundamenta a opção pelo balun choke de ar e apresenta sua caracterização experimental.

2 Efeito Pelicular e Irradiação Eletromagnética

O efeito pelicular ocorre em condutores submetidos a correntes alternadas de alta frequência, como 868 MHz, fazendo com que a corrente se concentre majoritariamente em uma fina camada na superfície do condutor, reduzindo a profundidade de penetração (δ) disponível para o transporte de corrente. A profundidade de pele é dada por:

$$\delta = \sqrt{\frac{2\rho}{\omega\mu}}, \tag{1}$$

onde ρ é a resistividade do material condutor, $\omega = 2\pi f$ é a frequência angular e μ é a permeabilidade magnética do material.

Para os materiais usados neste projeto:

- **Cobre (usado no RG174):** $\rho_{\text{Cu}} \approx 1.68 \times 10^{-8}\ \Omega \cdot \text{m}$;
- **Latão (brass, usado nos elementos da antena):** $\rho_{\text{brass}} \approx 6.2 \times 10^{-8}\ \Omega \cdot \text{m}$.

O cálculo para cobre, em 868 MHz, resulta em:

$$\delta_{\text{Cu}} \approx \sqrt{\frac{2 \times 1.68 \times 10^{-8}}{2\pi \times 868 \times 10^{6} \times 4\pi \times 10^{-7}}} \approx 2.2\,\mu\text{m} \tag{2}$$

e, para o latão, δ_{brass} será ligeiramente maior devido à sua maior resistividade.

Essa profundidade reduzida faz com que tanto o RG174 quanto os elementos de latão da antena concentrem as correntes de RF em uma camada superficial muito fina, aumentando sua sensibilidade a perdas superficiais e imperfeições. No caso do cabo, o efeito pelicular atua predominantemente sobre o cobre (ou cobre estanhado) da malha e do condutor central, enquanto na antena o efeito é acentuado pelo uso do latão.

A combinação das propriedades de materiais e geometrias do sistema pode favorecer o surgimento de correntes de modo comum. Essas correntes fluem na mesma direção tanto pelo condutor central quanto pela parte externa da malha do coaxial, ao invés de se cancelarem (caso ideal do modo diferencial). O resultado é que a malha do cabo atua como uma antena secundária ou espúria, irradiando energia e distorcendo o padrão de radiação, além de modificar a impedância vista pelo transmissor ([3]; [4]; [5]).

Ressalte-se que, em 868 MHz ($\lambda \approx 34{,}5$ cm), qualquer assimetria de fixação, diferença de material, ou desbalanceamento de montagem pode ser significativa em relação ao comprimento de onda, agravando a indução de correntes parasitas mesmo diante de pequenas imperfeições construtivas.

Por esse motivo, é fundamental considerar as características dos materiais — tanto do cabo RG174 quanto dos elementos de latão — para o correto entendimento do efeito pelicular e do potencial de irradiação indesejada em medições de antenas em altas frequências.

3 Uso de Balun Choke para Supressão de Correntes de Modo Comum

O uso de linhas de transmissão coaxiais para alimentar antenas simétricas, como dipolos, pode introduzir correntes de modo comum devido a diferenças geométricas ou de materiais entre os lados da antena e à natureza do acoplamento entre antena e cabo ([1]). Essas correntes, intensificadas pelo efeito pelicular, induzem irradiação secundária a partir do

cabo, distorcendo o padrão de radiação e a impedância observada (2). Para mitigar esse problema, a literatura recomenda o uso de baluns (*BALanced to UNbalanced transformers*), que garantem que apenas a corrente diferencial alimente a antena, suprimindo o modo comum (6; 3).

Diversos tipos de baluns existem, incluindo baluns de tensão, de corrente e transformadores de impedância. O balun choke (*current balun*) destaca-se pela simplicidade, baixa perda e eficácia na supressão de correntes de modo comum, sem alterar a impedância (7; 8). Embora baluns comerciais, baseados em ferrites ou núcleos toroidais, estejam disponíveis em fornecedores como Mini-Circuits e DX Engineering (9), a fabricação própria de um balun choke de ar permite ajuste preciso dos parâmetros construtivos, flexibilidade para experimentação e redução de custos (3).

Neste projeto, foi fabricado um balun choke de ar com as seguintes especificações:

- Frequência de operação: 868 MHz;
- Número de espiras: 7;
- Diâmetro do enrolamento: 24 mm;
- Cabo: coaxial RG174 (diâmetro de 2,6 mm, condutor central de cobre nu, malha de cobre estanhado);
- Estrutura: suporte cilíndrico impresso em 3D.

A indutância foi projetada para fornecer alta reatância indutiva a 868 MHz, conforme as fórmulas de dimensionamento de solenóides (7; 6). A validação experimental foi realizada com um analisador vetorial de redes (VNA), descrita a seguir.

3.1 Resultados Experimentais do Balun Choke Fabricado

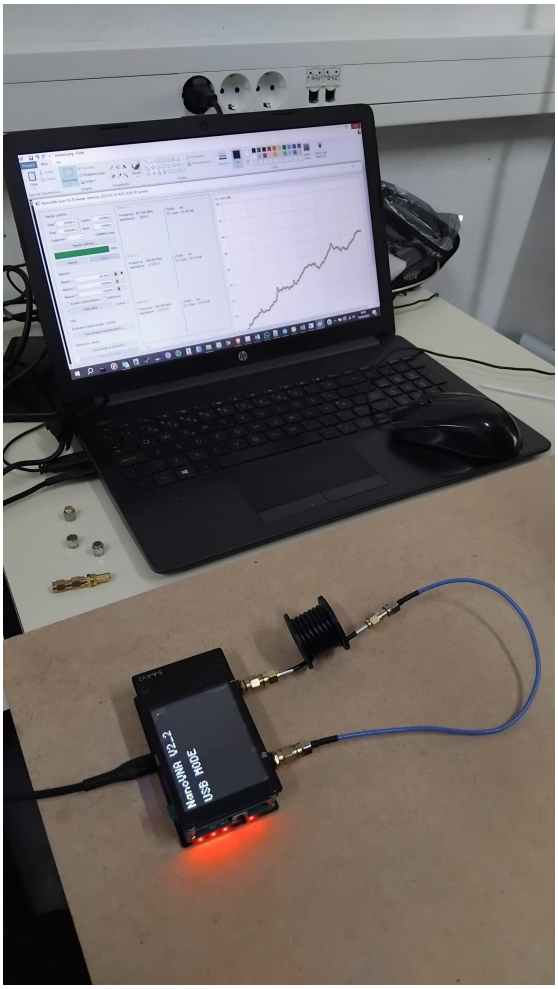
Para validar o desempenho do balun choke, foram realizadas medições de S21 no VNA em duas configurações:

(a) **Modo diferencial (conexão normal):** Medido S21 com sinal aplicado centro a centro e malha a malha (modo útil).

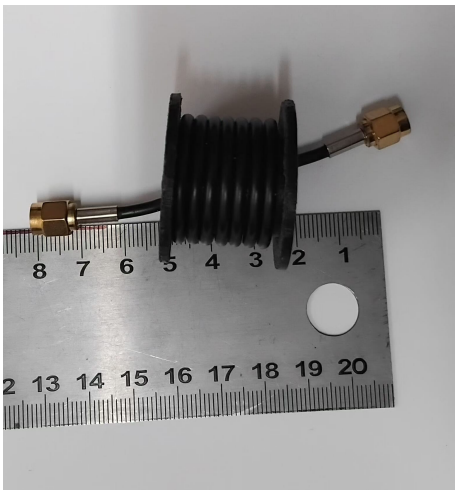
(b) **Modo comum (apenas malha, pino central interrompido):** Medido S21 simulando corrente de modo comum, aplicando sinal apenas entre as malhas.

As figuras a seguir ilustram o setup experimental, o balun choke de ar fabricado e os resultados das medições no VNA. A Figura 1 apresenta, à esquerda, o balun durante a caracterização (1a), e à direita, fotos do balun com régua para escala (1b) e uma vista superior do dispositivo (1c). Já a Figura 4 mostra os gráficos das respostas S21 obtidas no VNA, sendo que o subgrafo 4a corresponde ao modo diferencial (sinal normal do cabo) e o subgrafo 4b ao modo comum (corrente indesejada na superfície externa da malha), ambos medidos na faixa de 300 MHz a 1 GHz.

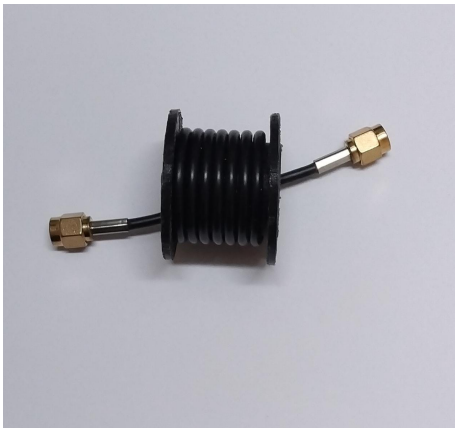
Os resultados confirmam que o balun choke de ar apresenta baixíssima perda no modo diferencial (menor que $-0,6$ dB) e excelente supressão de modo comum (atenuação de -47 dB ou mais em 868 MHz). Essa performance se compara favoravelmente a baluns comerciais para a mesma faixa de frequência, validando o projeto e destacando a viabilidade da solução de baixo custo para medições de antenas.



(a) Balun durante caracterização.

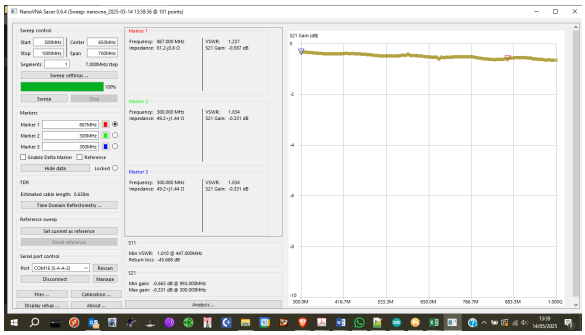


(b) Medida do balun com régua.

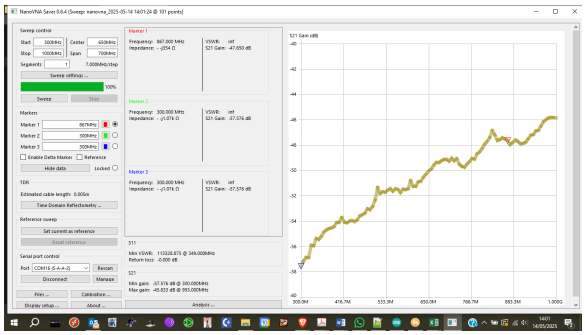


(c) Vista superior do balun.

Figure 1: Fotos do balun choke de ar fabricado usando cabo RG174, com suporte impresso em 3D.



(a) Modo diferencial: perda de inserção de $\approx -0,58$ dB em 868 MHz.



(b) Modo comum: atenuação superior a -47 dB em 868 MHz.

Figure 2: Resultados experimentais de S21 do balun choke de ar fabricado, obtidos em analisador vetorial de redes (VNA) na configuração normal (diferencial) e na configuração de modo comum.

3.2 Análise do Dimensionamento do Balun Choke

O dimensionamento do balun choke foi realizado com um script MATLAB, baseado na fórmula de indutância para uma bobina de camada única ([10]; [7]). A indutância alvo foi definida como 550 nH, gerando uma reatância de:

$$X_L = 2\pi \times 868 \times 10^6 \times 550 \times 10^{-9} \approx 3000\ \Omega, \tag{3}$$

suficiente para bloquear correntes de modo comum. O script calcula a indutância (L_s) em função do número de espiras (N) para o cabo RG174 ($d_{\text{cabo}} = 2.6$ mm) e diâmetro do enrolamento ($D = 2.4$ cm):

```
% Dimensionamento de um Balun de estrangulamento
% cabo coaxial RG174 (2.6 mm de diâmetro)
% Fórmula da indutância para uma bobina de camada única:
%       4 pi^2 N^2 r^2
% Ls = ----------------- x K
%             l
% l = N * dcabo; % comprimento da bobina
% K = f(l/D) onde D = 2r
% https://coil32.net/single-layer-coil.html
Lt = 550; % Indutância alvo de 550 nH
dcabo = 0.26; % Diâmetro do cabo em cm (2,6 mm)
N = 1:10; % Número de espiras testadas (1 a 10)
l = N * dcabo; % Comprimento da bobina (cm)
D = 1.8; % Diâmetro interno da bobina em cm
L = 2 * pi * D .* N.^2 .* (log(1 + (pi * D) ./ (2 * l)) + 1 ./
    (2.3004 + 3.437 * (l ./ D) + 1.7636 * (l ./ D).^2 -
    (0.47 ./ (0.755 + D ./ l).^1.44)));
plot(N, L, N, ones(size(N)) * Lt)
xlabel('Número de espiras');
ylabel('Indutância/nH');
grid on
```

O script sugere que $N = 7$ espiras atinge a indutância alvo. O diâmetro do enrolamento foi ajustado para $D = 2.4$ cm (24 mm), conforme especificado no projeto. **Recomenda-se verificar a frequência de autorressonância (f_r) para garantir $f_r > 868$ MHz.**

4 Casamento de Impedâncias

O casamento de impedâncias é um aspecto crucial para a otimização da transferência de potência entre o transmissor e a antena Yagi-Uda, originalmente projetada com uma impedância de 30 ohms. Este ajuste é fundamental para compatibilizar com a linha de transmissão de 50 ohms.

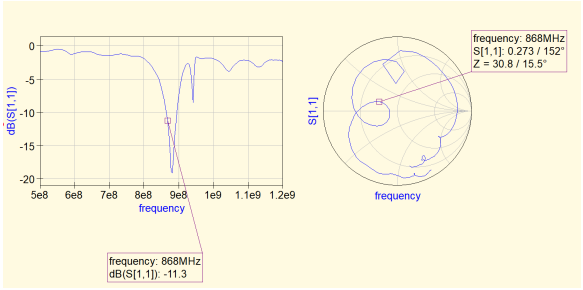
4.1 Importância do Casamento de Impedâncias

Um casamento eficiente garante que a potência do transmissor seja maximamente transferida para a antena, minimizando assim as perdas por reflexão. Desbalanceamentos e

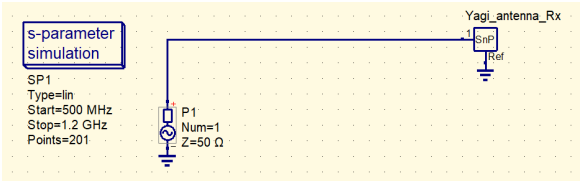
reflexões impactam diretamente nas medições das antenas, podendo distorcer o padrão de radiação e comprometer a confiabilidade dos resultados. Sistemas mal casados também resultam em maior potência refletida, o que pode causar aquecimento e possíveis danos ao transmissor.

4.2 Metodologia e Ferramentas Utilizadas

Para validar o processo de casamento de impedâncias, foram utilizados os softwares Qucs e SimNEC. O Qucs permitiu a simulação e cálculo dos componentes necessários para o circuito de acoplamento de impedância, enquanto o SimNEC possibilitou visualizar as melhorias antes e após o casamento.

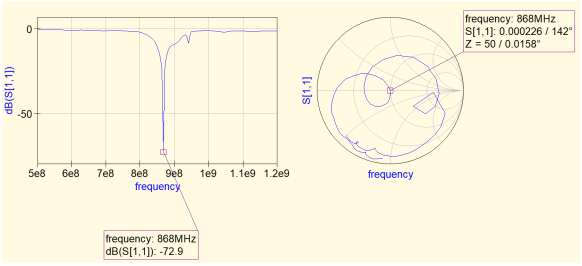


(a) Resultado.

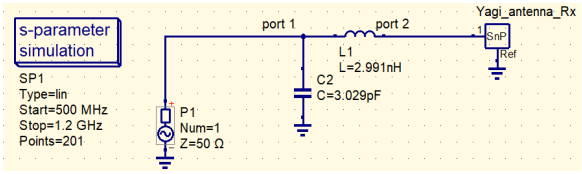


(b) Circuito.

Figure 3: Representação S11 da antena Yagi-Uda antes do casamento de impedâncias.



(a) Resultado.



(b) Circuito.

Figure 4: Representação S11 da antena Yagi-Uda depois do casamento de impedâncias.

A placa de circuito impresso para o casamento de impedância foi projetada com valores calculados no Qucs, como mostrado na Figura 5 e desenhada no EasyEDA, com previsão de produção em CNC para maior precisão.

4.3 Resultados Esperados

Os resultados comparativos serão apresentados na seção seguinte. Com foco em avaliar o desempenho sem e com o uso do balun e da placa, busca-se comprovar a eficácia do ajuste.

Esta abordagem garante eficiência do sistema e que as medições reflitam o verdadeiro desempenho da antena.

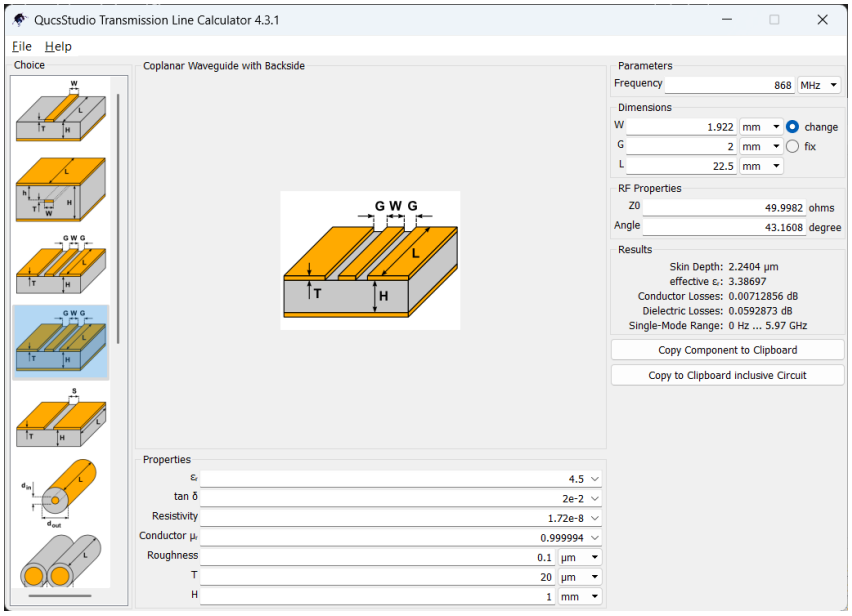


Figure 5: Dimensionamento da placa para o casamento de impedâncias.

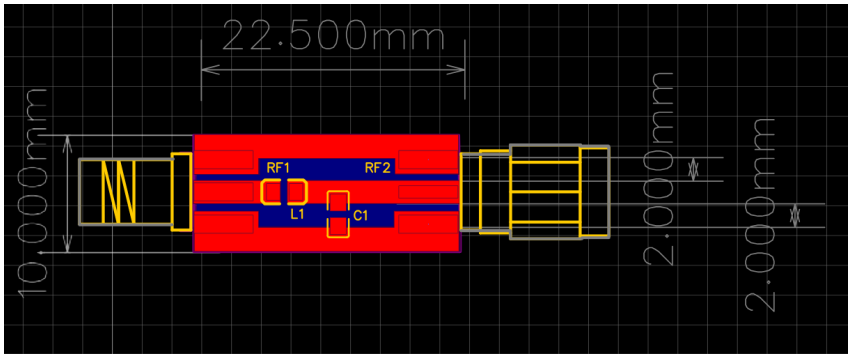


Figure 6: Dimensionamento da placa no EasyEDA

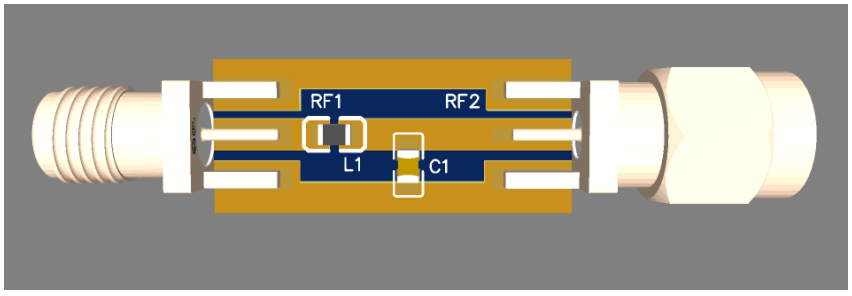


Figure 7: Placa 3D no EasyEDA

5 Conclusão

O balun choke de ar projetado para 868 MHz elimina correntes de modo comum, suprimindo a irradiação eletromagnética indesejada causada pelo efeito pelicular. Validações experimentais confirmam seu desempenho, com atenuação de -47 dB no modo comum e baixa perda no modo diferencial. O dimensionamento, apoiado por cálculos e simulações, assegura medições precisas do padrão de radiação. Futuras etapas incluem a comparação de medições com e sem o balun e o projeto de redes de acoplamento de impedância para

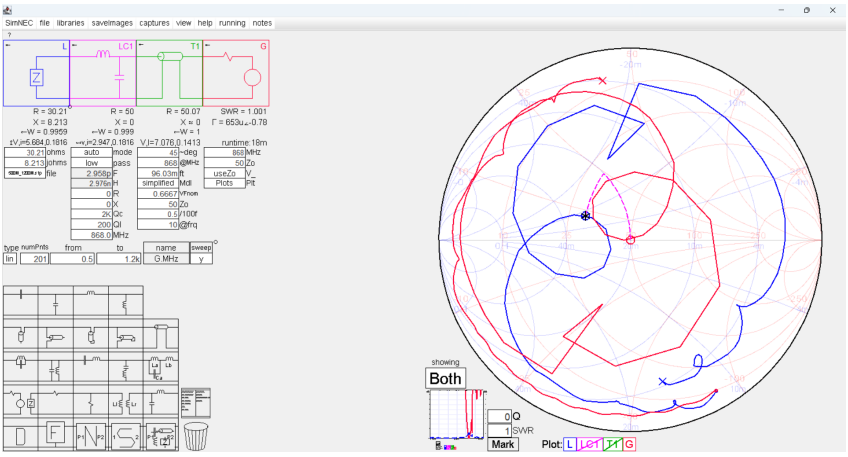


Figure 8: Simulação do SimNEC mostrando melhora no ajuste de impedância.

otimizar a transferência de potência.

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Two-Axis Computerised Antenna Radiation Pattern Measurement System

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Abstract. Hands-on experimentation is essential in RF and telecommunications education, yet access to instrumentation for characterising antenna performance, such as radiation patterns, is often limited due to the high cost and proprietary nature of commercial systems. While simulation software can estimate patterns during design, real-world environmental factors significantly affect performance. In this reference frame, this work presents the design and implementation of a custom-made mechatronic system able to measure antenna radiation patterns. By allowing real-time visualisation, this test rig promotes a deeper understanding of the antenna's behaviour in a classroom context. The device, which besides educational can also be used for research purposes, features two degrees of freedom, supports various antenna types, and has been validated using a custom-built 868 MHz Yagi-Uda antenna. Experimental results closely match the ones obtained from simulation software, and all design files and code will be publicly released to support replication and community-driven development.

Keywords: Antenna Radiation Pattern, Measurement System, Educational Methodologies, Mechatronics

1 Introduction

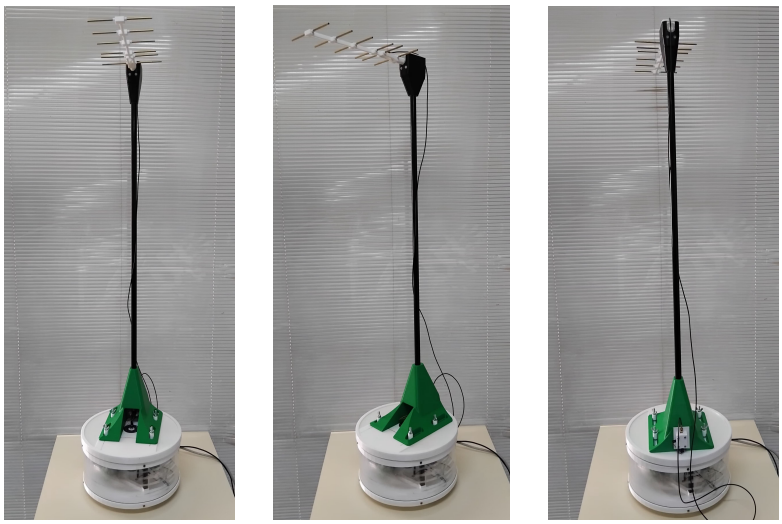
Characterisation of antenna radiation patterns is a fundamental aspect of wireless communication engineering, since the spatial distribution of radiated energy directly affects coverage, RF interference, and overall system performance [1]. While electromagnetic simulation tools are commonly employed to provide theoretical estimates of these patterns during antenna design, real-world measurements are indispensable to capture the effects caused by environmental factors and manufacturing tolerances, which simulations cannot fully replicate.

For those reasons, having access to measurement systems for antenna radiation patterns is a valuable tool for both research, product development and education applications. Regarding the latter, the use of such measurement setups in educational environments remains limited due to factors such as high acquisition costs, physical complexity, and frequently the need for specialised

infrastructure such as anechoic chambers. These limitations often restrict the ability of students to engage in hands-on experimentation, confining learning primarily to simulations and theoretical concepts. This gap can make the understanding of electromagnetic phenomena harder and reduce opportunities for experiential problem-solving.

To address these challenges, there has been growing interest in developing practical teaching tools. Various laboratory solutions and open-source projects have been proposed, aiming to reduce costs and complexity. Nevertheless, many of these approaches concentrate on specific frequency bands or lack the flexibility needed for broad didactic applications. Consequently, there remains a demand for accessible, modular, and low-cost platforms that can support experimental activities while illustrating the main antenna behaviour concepts.

In this context, the present work introduces the design and implementation of a computerised, two-axis radiation pattern measurement system, primarily intended to be used in teaching activities at the Polytechnic Institute of Bragança. The mechanical aspect of this measuring device is shown in Figure 1.



(a) Front view (b) Side view (c) Back view.

Fig. 1: The antenna radiation pattern measurement system.

The details behind the construction of this equipment will be presented later in Section 3. But first, an overview of existing similar measurement systems will be provided in Section 2, where both advantages and limitations of each solution will be highlighted. The result of this review and preliminary analysis allows this work to be positioned within the panorama of existing solutions. Section 4 will provide the experimental results obtained using a proof antenna. The close proximity between the experimental results and those generated by electromagnetic

simulation software will become evident, which validates the current approach. Finally, Section Finally, Section 5 will present the final conclusions of this work as well as some guidelines that will be followed in its future development.

2 State of the Art

The fundamental importance of antenna radiation pattern characterisation extends not only to wireless communication system optimisation but also to engineering education, where understanding the way those patterns emerge is critical to understanding signal propagation and antenna performance. However, the abstract nature of these electromagnetic concepts often makes them difficult to grasp, especially in the absence of practical visualisation tools. Those facts highlight the need for accessible experimental platforms that aid student learning by linking theory and practice.

Commercial antenna radiation pattern measurement systems, exemplified by devices such as the MegiQ RMS-0460 and Diamond Engineering D6050, are primarily designed for research and industrial applications. While they offer high accuracy and advanced features, their significant acquisition cost, operational complexity, and infrastructural requirements generally hinder their adoption within educational institutions that prioritise simplicity and affordability.

In order to overcome these barriers, various custom and open-source solutions have been presented in the literature. For example, Picco and Martin [2] proposed a low-cost setup based on modified Wi-Fi hardware and open-source firmware capable of measuring RF signals at 2.4 GHz. Their proposal includes an automated antenna positioning using stepper motors integrated with LabVIEW software.

Junfithrana et al.[3] and Simanjuntak et al.[4] describe the development of automated two-axis measurement systems controlled by microcontrollers and integrated with network analysers and Labview. Their aim was to demonstrate how accessible electronic components can be integrated into a measurement apparatus for educational purposes.

Additionally, open-source and software-defined radio (SDR) based platforms have broadened the scope for flexibility and adaptability in academic environments. Projects such as those in the Weber State University leverage SDRs, Arduino development boards, and GNU Radio to promote accessible measurement capabilities [5, 6]. Hardware improvements, including substituting PVC with 3d-printed parts and adopting modular assemblies, have contributed to enhanced portability and ease of customisation. Concurrent software advancements focusing on data processing, noise reduction, and error mitigation [7] support more reliable experimental outcomes and collaborative student research across diverse fields.

All of these works served as inspiration, but none of them met the design criteria that we would like a system of this nature, completely open-source, to have. Indeed, this work builds upon this foundation, offering a low-cost, computerised two-axis measurement system that complements theoretical instruction

with real-world experimental interaction. By facilitating interactive visualisation and analysis of antenna radiation patterns, the system provides an alternative pedagogical resource suitable for a variety of teaching and learning scenarios.

In the remaining sections of this paper, the antenna radiation measurement system will be explained, taking into consideration its distinct features with special focus on its mechatronic details.

3 Materials and Methods

As discussed previously, commercial antenna radiation pattern measurement systems, while highly accurate, are often prohibitively expensive and complex for educational environments. To address these challenges, a low-cost, replicable system was developed to enable didactic and experimental measurement of antenna radiation patterns. This section outlines the materials used, system configuration, and the methods adopted for calibration and validation.

3.1 System Description

The measurement system developed employs a structure with two rotational axes, enabling automatic scanning of the radiation pattern for antennas mounted on its upper support. The azimuth axis, located at the base, allows rotation from 0 to 360° in the horizontal plane. Above it, a perpendicular axis enables rotation in the orthogonal plane, also covering up to 360°. Depending on the antenna mounting orientation, this perpendicular axis may be adapted to scan the elevation axis. This configuration provides flexible and computer-controlled angular positioning of the antenna, allowing for antenna characterisation from different perspectives.

The system architecture is modular and structured into three primary subsystems: mechanical, electrical and software. The former includes all the rotational assemblies, supports, brackets, and transmission mechanisms. The transmission mechanisms, comprised of gears and drive belts, are propelled by NEMA 17 stepper motors. The electric signals used to move those motors are generated by a power driver, in this case a DRV8825, controlled through an Arduino Uno development board. The electrical subsystem also includes Hall-effect sensors for global system positioning and an RF power sensor based on the AD8313 integrated circuit. The third subsystem includes the firmware that lives in the microcontroller and the graphical interface developed for MATLAB[3]. Each one of those three subsystems will be further detailed in the next sections.

3.2 Mechanical Subsystem

From a mechanical point of view, the system can be seen as consisting of three parts. An upper part, responsible for holding the antenna, and able to tilt along

[3] A campus version of this software is licensed to the Polytechnic Institute of Bragança.

its axis. Figure 2 illustrates this part modelled in SolidWorks[4] CAD software. The two distinct rotational movements are highlighted: azimuth (blue arrow) and perpendicular rotation (red arrow). The transmission mechanism inside this part has been specially designed so as not to include any metal parts. In fact, all of the existing gears and bearings that support the main shaft are made of nylon. This aims to minimise any pattern distortion that can be caused by metallic parts close to the antenna.

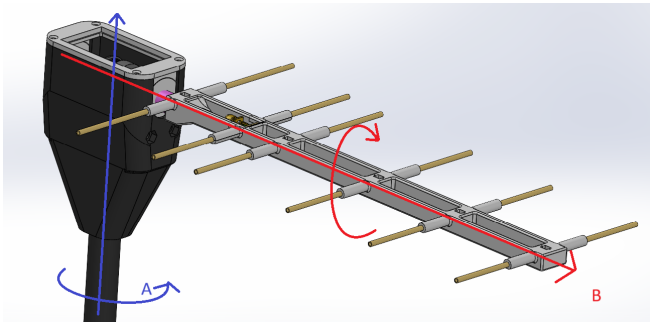


Fig. 2: SolidWorks rendering of the antenna measurement system highlighting the rotational movements: (A) azimuth rotation and (B) perpendicular axis rotation.

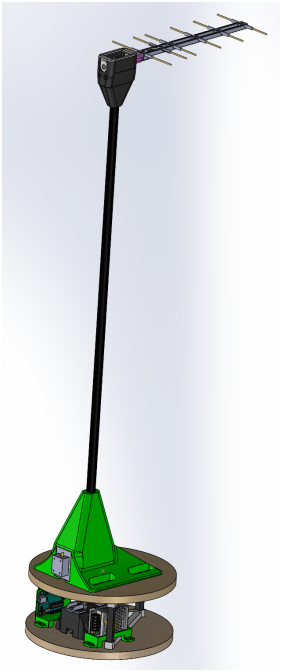
It is worth noticing that, even if Figure 2 depicts a particular type of antenna, the platform is intended for experiments with various antenna types. A quick-change bracket was designed to enable the fast replacement of antennas.

After this top antenna handler part, follows a central part consisting of a one-meter hollow fibreglass tube and a mechanical transmission system. The material used for the tube was carefully chosen to be non-conductive but, at the same time, robust and resilient to harsh environmental conditions. The tube length was selected such that it corresponds to three wavelengths for the band of interest.

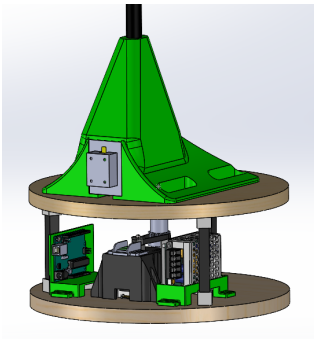
The tilt mechanism on the top part is driven by pulleys and a belt that runs inside the fibreglass tube. The mechanical actuation is provided by a stepper motor housed in the central part. At the same time, this central part, which is suspended by a system of ball bearings, is able to rotate, propelled by a gear mechanism located at the bottom part. This bottom part, where all the main electronic parts are housed, is made from wood-based chipboards cut in a circular shape with a 30 cm diameter. Figure 3 illustrates each one of the three mechanical parts.

Many of the fittings and stands were CAD-designed and 3d printed using polylactic acid (PLA).

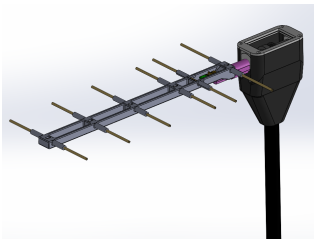
[4] Educational version licensed to the Polytechnic Institute of Bragança.



(a) Full system.



(b) Bottom section.



(c) Top section with test antenna.

Fig. 3: SolidWorks rendering of the antenna radiation pattern measurement system.

3.3 Electrical Subsystem

This section deals with the electrical subsystem, which is illustrated by the block diagram of Figure 4.

All the electrical power required to drive the system is provided by a 12 V, 2 A, AC to DC switch-mode power supply. The output voltage is connected to the local linear voltage regulator in the Arduino board and to the stepper motor driver. The stepper driver is based on the DRV8825 integrated circuit which is able to operate with voltages that span from 8 V to 45 V and is capable of driving up to 2.5 A for each of its two outputs. Among its many features, it includes several protection mechanisms, such as overcurrent and thermal shutdown, as well as the possibility of operating directly with microsteps. This driver is used for open-loop control of two NEMA 17 stepper motors. Those are two-phase motors presenting a 1.8 ° step angle and around 0.5 Nm of holding torque. The reference position of each stepper motor is provided by a Hall-effect sensor, which detects the magnetic field of a permanent magnet coupled to each motor shaft.

Finally, the RF power harvested by the antenna (DUT) is measured by the Analog Device's AD8313 chip. This integrated circuit behaves as a multistage demodulating logarithmic amplifier and is able to generate a DC voltage propor-

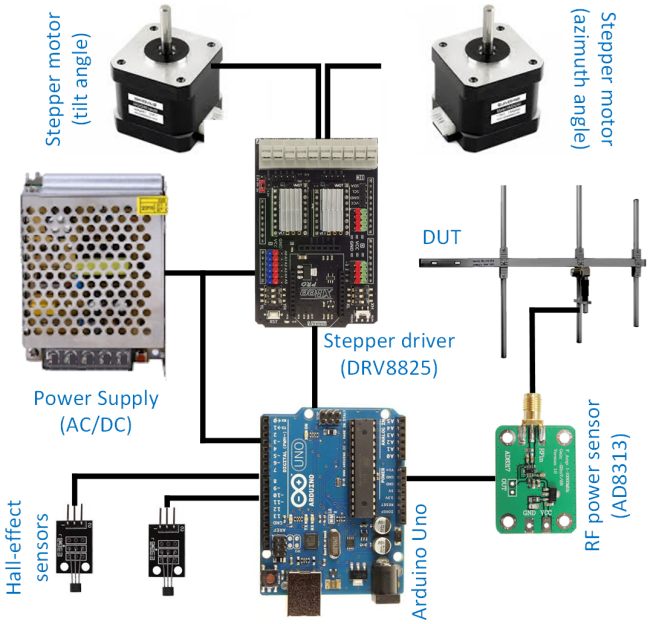


Fig. 4: Block diagram of the electronic modules and their connection.

tional to the input RF power. This device has a bandwidth that extends from 100 MHz to 2.5 GHz with a total dynamic range of 70 dB.

3.4 Software Subsystem

System control and automation are achieved through embedded firmware running on the Arduino UNO platform. The firmware, developed in C++, controls the movement of both stepper motors (azimuth and perpendicular axes), executes homing routines, and gathers and filters data delivered by the AD8313 sensor.

At a higher level, a MATLAB program takes care of the data acquisition process and user interface. Communication between MATLAB and the microcontroller is established through a serial connection over USB. MATLAB sends motion commands and data requests to the microcontroller. The software application also handles homing routines and manages retransmission in case of communication errors. Acquired data are stored and immediately presented as polar plots, enabling straightforward visualisation of radiation patterns for analysis and educational use.

3.5 System Operation

When initialised, the system begins by running a homing calibration. During this stage, Hall-effect sensors allow the system to automatically detect the reference

point (zero) for each axis. The stepper motor's driver was set to operate in micro-stepping mode (1/16). A transmitting antenna is operating as a beacon, transmitting at a given frequency.

Data provided by the RF power sensor is acquired at a user-defined angular interval and sent to MATLAB to be processed and visualised. The entire measurement and data acquisition flow, along with system interconnections, is summarised by Figure 5.

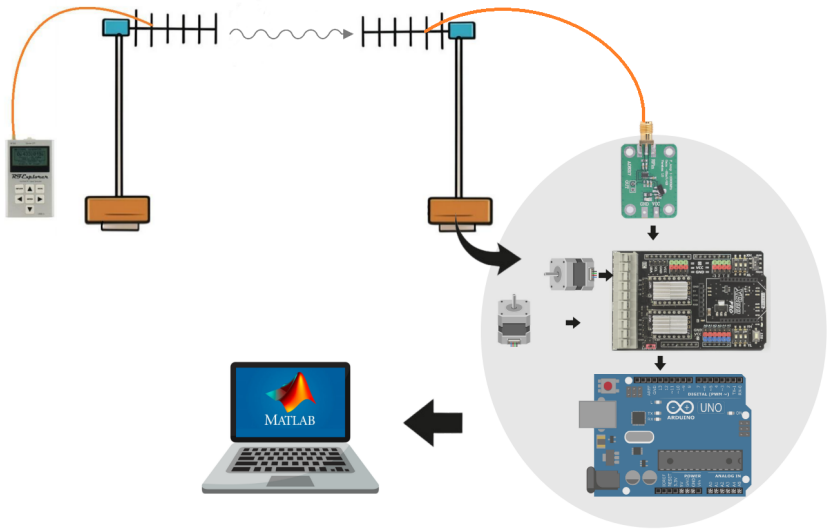


Fig. 5: Schematic showing the flow of the signal from generation to analysis.

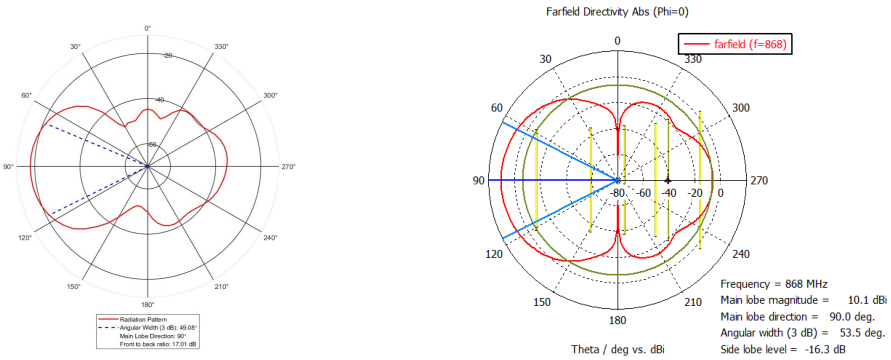
4 Results and Discussion

To validate the antenna radiation pattern measurement system, a Yagi-Uda antenna, to operate at 868 MHz, was sized and built. During this stage, simulation was performed using CST Studio Suite[5]. Those simulations would serve as a term of comparison for the results obtained experimentally.

Experimental measurements were performed along the azimuthal plane with a 5 ° resolution. Figure 6 presents a comparison between the measured radiation pattern and the simulated pattern obtained. The measured data, shown in Figure 6a, were obtained by averaging 100 samples per angular interval. The simulated pattern, depicted in Figure 6b, serves as a theoretical reference for validation.

As shown in Figure 6, both the measured and simulated patterns exhibit a main lobe direction at 90°, demonstrating that the system is able to determine

[5] The free educational version of this software was used in the simulations.



(a) Real data acquired (b) Simulation of the radiation pattern

Fig. 6: Comparison of real and simulated radiation pattern.

the direction of the main lobe. The measured beamwidth (-3 dB) of 49.08° is closely aligned with the simulated value of 53.5°, front-to-back ratio is 17.01 dB while the simulated value is 16.31 dB, a comparison not shown in the figure. These results suggest that the system can capture the main features of the radiation pattern, including the main lobe and secondary lobes.

Some discrepancies were observed between the measured and simulated patterns, which can be attributed to several factors. The measurement environment, although selected to minimise reflective effects, could not eliminate all sources of interference, unlike the free-space conditions assumed in the simulation. Additionally, the antenna may have captured signals at frequencies other than the intended 868 MHz, and the presence of other transmitting devices in the vicinity likely introduced noise and interference. Such effects could be mitigated in a controlled setting, such as an anechoic chamber, which was not available for this study. These environmental factors highlight the challenges of real-world measurements compared to the idealised conditions of the simulation, yet the system's ability to produce consistent and reproducible results under these conditions demonstrates its robustness for educational purposes.

Overall, the results demonstrate that the system is capable of capturing key radiation pattern features, with measurements that generally agree with the simulation. While environmental limitations affected absolute accuracy, the system remains a robust and versatile platform for practical education in telecommunications. These results support the suitability of the system as an alternative learning resource for exploring fundamental concepts in antenna behaviour, especially in academic and training environments where cost is a limiting factor.

5 Conclusions and Future Work

In the course of this article, the design, implementation and testing of a system that allows the measurement of the radiation pattern of antennas was docu-

mented. Although these types of devices exist commercially, their price makes them quite inaccessible for use in an educational context.

The designed system has two degrees of freedom, allowing the automatic acquisition of the radiation pattern of an antenna in 360 degrees.

Preliminary tests carried out along the azimuthal plane suggest that the presented methodology provides a close match to those obtained by simulation.

Future improvements include the addition of bandpass filters to suppress out-of-band signals, the addition of a balun to mitigate transmission line imbalance, and the improvement of impedance matching.

Additional trials using different antenna types and along different planes will be performed. Once thoroughly tested, the entire project will be placed in a repository to promote its replication. Moreover, interactive learning modules, aiming to increase the educational engagement with the subject of RF engineering and antenna theory, will be developed.

Acknowledgment

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Two-Axis Computerized Antenna Radiation Pattern Measurement System

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Trabalho realizado sob a orientação de

Prof. Dr. João Paulo Coelho

Licenciatura em Engenharia Eletrotécnica e de Computadores

2024-2025

Two-Axis Computerized Antenna Radiation Pattern Measurement System

Relatório da UC de Projeto

Licenciatura em Engenharia Eletrotécnica e de Computadores

Escola Superior de Tecnologia e Gestão

Mauro da Silva Leme - 52965

2024-2025

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Mauro da Silva Leme - 52965

Dedicatória

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Agradecimentos

(agradecimentos)

Resumo

Compreender o padrão de radiação de uma antena é crucial para otimizar sistemas de comunicação sem fio. O padrão de radiação revela como uma antena transmite ou recebe ondas eletromagnéticas em diferentes direções, ajudando a determinar sua área de cobertura, intensidade do sinal e possível interferência com outros dispositivos. Este conhecimento garante uma distribuição eficiente do sinal, minimiza o desperdício de energia e melhora o desempenho geral do sistema ao permitir o posicionamento e orientação precisos das antenas. O padrão de radiação de uma antena pode ser estimado durante o processo de design por meio de software EMS. No entanto, na prática, o padrão de radiação de uma antena pode diferir significativamente das simulações devido à influência do ambiente onde está instalada.

Enquanto as simulações geralmente assumem condições ideais e controladas, fatores do mundo real, como edifícios próximos, terreno, condições meteorológicas e outras antenas, podem causar reflexões, difrações e absorções de sinal que alteram o padrão de radiação. Estruturas metálicas, por exemplo, podem criar interferências ou efeitos de multipercurso, enquanto obstáculos como árvores ou paredes podem atenuar sinais, deslocando a cobertura idealizada. Além disso, os efeitos do solo e a altura em que a antena é montada podem alterar drasticamente o padrão. Compreender esses impactos ambientais é essencial para prever com precisão o desempenho da antena em campo e garantir uma comunicação confiável em aplicações práticas.

Tipicamente, o equipamento para medir o padrão de radiação de antenas, pelo menos com qualidade metrológica, é muito caro e pode até ser excessivo se o objetivo de seu uso for apenas educacional. Nesse contexto, este trabalho visa desenvolver um setup de baixo

custo para medir o padrão de radiação de antenas que operam na banda ISM (inferior a 3 GHz).

Mais tarde ainda vou colocar aqui um parágrafo sobre os resultados obtidos

Palavras-chave: Padrão de Radiação de Antenas, Comunicação Sem Fio, Banda ISM, Medição de Baixo Custo, Desempenho de Antenas..

Abstract

Understanding an antenna's radiation pattern is crucial for optimizing wireless communication systems. The radiation pattern reveals how an antenna transmits or receives electromagnetic waves in different directions, helping to determine its coverage area, signal strength, and potential interference with other devices. This knowledge ensures efficient signal distribution, minimizes wasted energy, and improves overall system performance by allowing precise placement and orientation of antennas. The radiation pattern of an antenna can be estimated during the design process through EMS software. However, in practice, an antenna's radiation pattern can differ significantly from simulations due to the influence of the environment where it is installed.

While simulations typically assume ideal controlled conditions, real-world factors such as nearby buildings, terrain, weather conditions, and other antennas can cause signal reflections, diffractions, and absorptions that alter the radiation pattern. Metal structures, for instance, can create interference or multipath effects, while obstacles like trees or walls can attenuate signals, shifting the idealized coverage. Additionally, ground effects and the height at which the antenna is mounted can drastically change the pattern. Understanding these environmental impacts is essential for accurately predicting antenna performance in the field and ensuring reliable communication in practical applications.

Typically, antenna radiation pattern equipment, at least with metrological quality, is very expensive and may even be overkill if the objective of its use is only educational. In this context, this work aims to develop a low-cost setup to measure the radiation pattern for antennas operating in the ISM band (lower than 3 GHz).

Mais tarde ainda vou colocar aqui um parágrafo sobre os resultados obtidos

Keywords: Antenna Radiation Pattern, Wireless Communication, ISM Band, Low-Cost Measurement, Antenna Performance.

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b

Chapter 1

Introduction

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1.1 Descrição do trabalho

blá blá...

1.2 Objetivos

blá blá...

1.3 Organização do relatório

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Chapter 2

State of the Art

2.1 Introduction

Over the last few decades, technology has advanced rapidly, especially in telecommunications. Antennas have a crucial role in this context as they are key elements in the communication chain, upon which the overall performance of the system strongly depends. Since World War I, the design and development of antennas has been a topic of great relevance, initially driven by military applications. Over time, this importance has increased with the introduction of new technologies, such as television, and continues to grow in the current era of mobile communications, including 5G and high-frequency Wi-Fi. The need to characterize and evaluate an antenna's performance has become even more crucial to ensure the efficiency and reliability of communication systems.

To ensure that antennas operate within expected parameters, it is essential to use measurement and calibration systems. These systems allow the evaluation of various antenna characteristics, including radiation pattern, gain, efficiency, and bandwidth. Measuring and validating antenna performance is vital to make sure they meet technical and regulatory specifications.

Radiation pattern measurement is a key step in the design process of antennas and wireless devices. The results of these measurements are used to validate simulated designs

and identify potential performance issues before final testing.

While commercially available antenna measurement systems are designed to provide precise and repeatable measurements, they are often expensive. Therefore, it is possible to develop a Custom-Made solution to measure specific antenna characteristics. These custom-made solutions offer a cost-effective alternative, addressing specific needs that commercial systems may not cover.

Some commercial solutions available on the market for the measurement of antenna radiation patterns will be presented, as well as research and development work that explores different technical approaches. The goal is to provide a theoretical and practical basis that supports the development of an effective and accurate solution, suited to the specific measurement needs, aligning with the purpose of this project.

2.2 Commercially Available Products

This section presents commercially available solutions for measuring antenna radiation patterns. These solutions are used in research and development laboratories as well as in companies that need to validate antenna performance across different frequencies. The following descriptions provide a detailed overview of the main features and the purpose for which each system is most suitable.

MegiQ RMS-0460

The MegiQ RMS-0460 system, manufactured in the Netherlands, is designed to perform antenna radiation pattern measurements in environments such as laboratories or offices, without requiring an anechoic chamber. It operates within a frequency range of 370 MHz to 6 GHz, as shown in Figure 2.1, providing 3-axis measurements in less than two minutes for a complete radiation pattern analysis. The system offers a measurement accuracy of ± 1 dB and repeatability of ± 0.5 dB. Depending on the configuration, the system can handle loads up to 7.5 kg or up to 30 kg, making it an option for wireless communication devices operating across multiple frequency bands. The approximate price is €16,950.



Figure 2.1: MegiQ RMS-0460 Measurement System.

Source: MegiQ, 2024.

D6050 Diamond Engineering

The D6050, produced by Diamond Engineering in the United States, is designed to perform measurements over a frequency range from DC up to 67 GHz, and up to 110 GHz when using waveguides. It offers an angular precision of 0.02° and supports loads up to 11.5 kg, making it suitable for testing larger and heavier antennas. The system, depicted in Figure 2.2, features a modular configuration that allows the addition of linear axes for automated measurement. Additionally, it includes software for control and data analysis, and it can operate in high-frequency measurements. The starting price is €12,880.



Figure 2.2: D6050 Measurement System.

Source: Diamond Engineering, 2024.

Microwave Factory Co., Ltd. (MWF)

Microwave Factory Co. Ltd., from Japan, offers an option for measuring antenna radiation patterns, operating in a range from 0.4 GHz up to 300 GHz, depending on the configuration. As shown in Figure 2.3, the system includes an antenna tower, azimuth positioner, elevator, RF absorbers, and control software (MATEOS.NET), enabling 2D radiation pattern measurements and antenna gain analysis. The system is customizable to accommodate equipment with varying weights, with a capacity of up to 1000 kg, which allows it to be used for large-scale projects in controlled environments. It offers an angular precision of 0.03° and repeatability of ±0.03°. However, it requires an anechoic chamber to ensure measurement accuracy. The total price for the complete system is approximately €368,000.

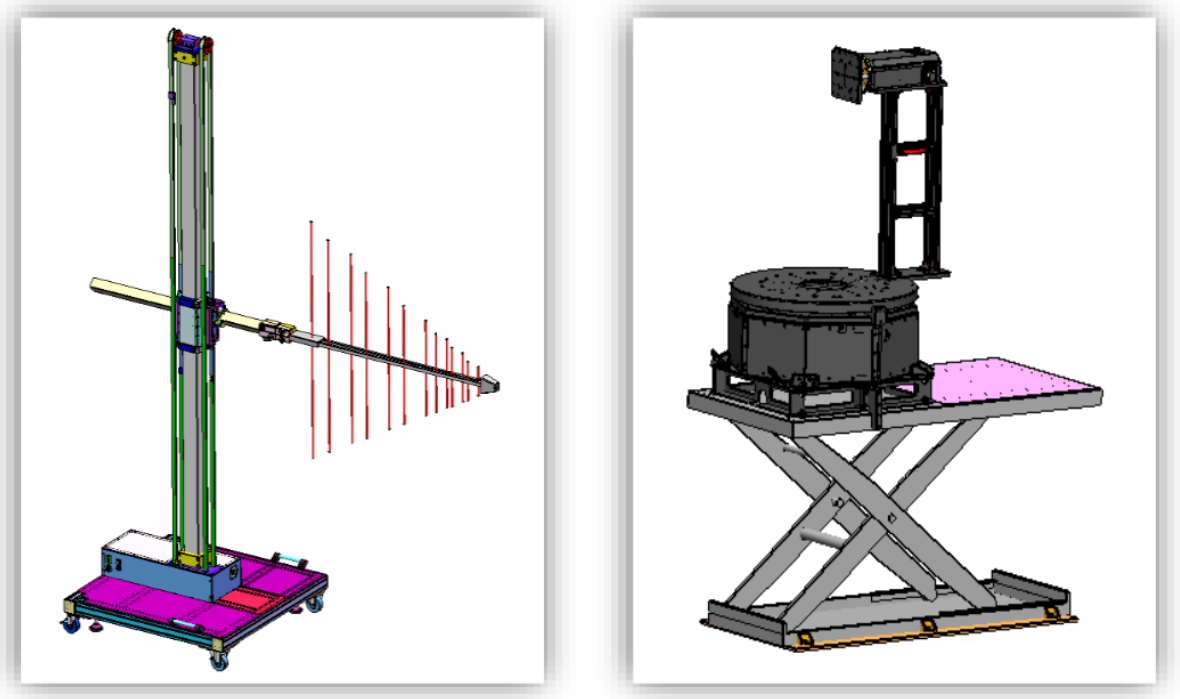


Figure 2.3: Microwave Factory (MWF) Measurement System representation.

Source: Microwave Factory Co. Ltd., 2024.

S7601A Antenna Measurement System

The S7601A Antenna Measurement System, developed by Keysight Technologies from United States, is a product for antenna characterization. It operates within a frequency range of 10 MHz to 50 GHz and it allows for both near-field and far-field measurements. This system supports 2-axis positioning and can handle antennas weighing up to 10 kg. It provides measurement accuracy up to ±0.8 dB, providing results for different antenna systems, including 5G, satellite, and radar applications. The system includes an integrated anechoic chamber, offering conditions for measurements without requiring an external chamber. Pricing for the S7601A is expected to be around €80,000, with exact pricing pending confirmation. The system is illustrated in Figure 2.4.(eu vou confirmar ainda esse preço).

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Figure 2.4: S7601A Antenna Measurement System.

Source: Keysight Technologies, 2024.

Contact was made via email with the companies that manufacture these systems. However, some specific information regarding certain features, such as signal accuracy or

load capacity, was either not provided or could be customized depending on the configuration of the system. This explains the gaps in the comparison table 2.1 below, where precise details were unavailable or subject to change based on client requirements.

Feature	**MegiQ RMS-0460**	**D6050 Diamond Eng.**	**Microwave Factory**	**S7601A**
Frequency Range	370 MHz – 6 GHz	DC – 67 GHz (110 GHz)	0.4 GHz – 6 GHz	10 MHz – 50 GHz
Measurement Axes	3-axis	2-axis	2-axis	2-axis
Signal Accuracy (dB)	±1 dB	Not Specified	Not Specified	±0.8 dB
Angular Precision (°)	Not specified	0.02°	0.03°	Customizable
Load Capacity	7.5 kg to 30 kg	11.5 kg	Up to 1000 kg	10 kg
Environment	No chamber required	No chamber required	Needs chamber	Chamber included
Price	€16,950	€12,880	€368,000	€856,000 (Chamber only)

Table 2.1: Comparison of Commercial Antenna Measurement Systems

The MegiQ RMS-0460 system is designed for smaller to medium-sized antennas, with a load capacity of up to 7.5 kg or 30 kg, depending on the configuration. It operates without requiring an anechoic chamber, making it applicable for environments like laboratories or educational settings, where portability and ease of setup are necessary.

The D6050 Diamond Engineering system provides an angular precision of 0.02°, suitable for applications that require accurate positioning of antennas. It operates within a frequency range up to 110 GHz when using waveguides, and supports high-frequency measurements for medium-sized antennas.

The Microwave Factory Co., Ltd. system has the highest load capacity among the compared systems, supporting up to 1000 kg. This makes it the choice for large-scale projects, such as those involving heavy or large antennas. It also provides a frequency range up to 300 GHz, making it suitable for advanced applications in aerospace or defense industries where high precision and anechoic chamber-controlled environments are critical.

The S7601A Antenna Measurement System has a load capacity of up to 10 kg and operates within a frequency range of 10 MHz to 50 GHz. It includes an integrated anechoic chamber, allowing measurements without the need for a separate chamber, making it applicable for industries such as telecommunications and satellite testing.

2.3 Research and Development Work

Based on the characteristics of the commercial equipment discussed above, it is possible to notice that while they are highly effective and precise, they can also be quite expensive. For many individuals, organizations, and companies, the high costs of these systems may not be justifiable, especially when specific measurements are needed, or financial resources are limited.

As a result, custom-made solutions emerge as an alternative. These solutions allow for the construction of customized measurement systems that can focus on specific antenna characteristics without the need to acquire complete and expensive commercial systems. Additionally, these solutions can be adapted to include functionalities not covered by commercial systems, providing greater flexibility and cost savings.

The existing literature currently demonstrates the viability and advantages of custom-made solutions for measuring antenna parameters, such as radiation patterns. These methods not only offer low-cost alternatives but also allow customizations that, in commercial systems, would be possible but with a significant increase in product costs. **Aqui vou indicar como realizei a pesquisa**.

Picco and Martin (2011)[1] developed "An Automated Antenna Measurement System Utilizing Wi-Fi Hardware". This work presents an innovative approach using commercial Wi-Fi hardware, specifically modified Linksys routers, to measure RF signals at 2.4 GHz. By replacing the original firmware with open-source DD-WRT firmware, they enabled the routers to function as cost-effective tools for measuring the received signal strength indicator (RSSI). Additionally, the system integrates with LabVIEW software to automate the positioning of the antenna under test using stepper motors. Although the system operates in a non-anechoic environment, it successfully minimizes inaccuracies, providing a viable low-cost solution for educational and research settings where the use of conventional equipment may be unavailable.

In Junfithrana *et al.* (2017)[2], "Development of Automated Antenna Radiation Pattern Measurement Using Rotator Application Model to Increase Accuracy", the authors

proposed an automated system that uses a two-axis rotator model controlled by microcontrollers. This set up is integrated with a network analyzer and the control software was developed in LabVIEW. The work highlights how the use of accessible components can increase measurement accuracy and reduce the time required to perform them, making the system a practical and economical alternative.

Simanjuntak *et al.* (2019)[3], presents the "Preliminary Design of Automatic Antenna Radiation Pattern Measurement System for Antenna and Propagation Laboratory Course", in which the authors develop an automated radiation pattern measurement system using microcontrollers. This study demonstrates that an accessible and portable system can be implemented in educational environments, providing quick and accurate measurements without the need for expensive commercial equipment.

These studies exemplify how custom solutions can be adapted to meet specific needs, offering an efficient alternative to commercial systems. They offer a solid foundation for developing new approaches that combine cost-effectiveness, flexibility, and innovation, aligning with the goal of this project to create an efficient, accessible, and adaptable antenna measurement system to meet the demands of the research environment.

Open-source solutions have gained prominence in the field of antenna measurement due to their flexibility, cost-effectiveness, and adaptability to different research and educational contexts. Examples of such approaches include the work developed by Weber State University, which proposed an antenna pattern measurement system using open-source components, as Software-Defined Radios (SDRs), Arduino microcontrollers, and GNU Radio software. These studies present the continuous evolution and improvement of accessible and highly effective antenna measurement systems.

In "Open-Source Antenna Pattern Measurement Validation Using FEKO" (2020)[4] presents the validation of an antenna pattern measurement system developed using SDRs and 3D-printed hardware. The system validation was conducted by comparing experimental measurements of antenna radiation patterns with theoretical simulations generated by the FEKO simulation software. The results demonstrated that the low-cost system could replicate measurements with a high degree of accuracy, making it an ideal tool for

undergraduate-level investigations on antenna theory and practical measurements.

In addition, the second article, "Open-Source Antenna Pattern Measurement System" (2020)[5], advances the development of the measurement system by replacing Wi-Fi routers with SDRs to significantly expand the frequency range (30 MHz to 6 GHz) over which measurements can be performed. This enhanced version of the system addresses challenges related to position control and measurement accuracy, using stepper motors controlled by an Arduino with open-source GRBL firmware. The study highlights how the replacement of components and the integration of new control and signal processing methods helped improve the system's precision and versatility while keeping it accessible and suitable for educational environments.

Finally, the most recent work among the studies reviewed, "Open-Source Antenna Pattern Measurement System: SDR-based Student Research and Development" (2022)[6], presents the latest updates to the system, including significant hardware and software improvements. Hardware upgrades include replacing PVC components with CAD-modeled and 3D-printed parts, increasing the system's mobility and adaptability for use in multiple noisy environments. The software was enhanced with new versions of Python, GNU Radio, and Linux, and advanced noise mitigation techniques and modulation control were incorporated to further improve measurement accuracy. The article highlights the system's success as a robust and flexible educational tool, creating opportunities for collaborative research and project development among electrical engineering students.

Chapter 3

Materials and Methods

Ainda vou incluir sobre os materiais para o casamento de impedancia e o balanceamento da linha de transmissão

Following the review of commercial systems and research initiatives discussed in the previous chapter, this chapter presents the practical implementation of a low-cost antenna radiation pattern measurement system. Designed to meet educational and experimental needs in the 868 MHz ISM band, the proposed setup combines mechanical design, custom electronics, and software integration. This chapter describes the materials used, the configuration of each subsystem, and the validation methodology. Additionally, it highlights RF-related improvements, including a custom-designed air-core choke balun and an impedance matching network, which significantly improve measurement reliability.

3.1 Proposed System Description

The antenna radiation pattern measurement system was developed with an emphasis on low cost, modularity, and suitability for educational and experimental applications. Built using commercially available components and custom-fabricated parts, the system allows for easy replication and future enhancements. It is composed of the following main subsystems:

- **Mechanical substructure** with two independent rotational axes (azimuth and perpendicular).
- **Electronic control system** based on the ATmega328P microcontroller (Arduino Uno).
- **RF power measurement circuit** using the AD8313 logarithmic detector.
- **Software suite** for motor control, data acquisition, and real-time visualization.

3.1.1 System Architecture

The system includes two rotational axes: the azimuth axis, located at the base, and a perpendicular axis at the top, which enables angular scanning in an orthogonal plane. Depending on the antenna mounting orientation, this perpendicular axis can be adapted to perform elevation sweeps. Both axes are actuated by NEMA 17 stepper motors with 1.8° step angle, driving belt-and-pulley mechanisms.

The transmission was specifically designed to achieve a resolution of approximately 1° per motor step, by using custom-designed gears with a 1.8:1 ratio. This configuration increases angular precision and mechanical torque, prioritizing measurement accuracy over rotational speed. The rotating assembly is supported by low-friction components to ensure smooth motion and reproducibility of measurements.

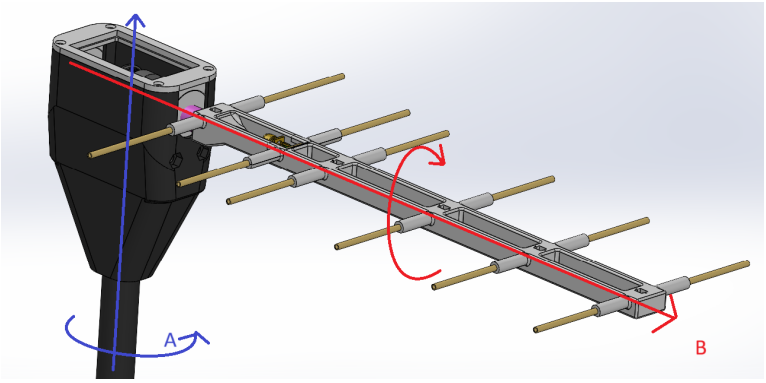


Figure 3.1: CAD model of the rotating platform with two degrees of freedom. Axis A corresponds to the azimuth rotation and axis B to the perpendicular rotation, which can be configured as the elevation axis.

3.1.2 Components Used

The main components used in the construction of the system are:

- **Stepper Motors**: NEMA 17 bipolar stepper motors with 1.8° per step, configured for 1/16 microstepping using DRV8825 drivers.
- **Microcontroller**: Arduino Uno board based on the ATmega328P, responsible for coordinating motor motion, executing homing routines, and reading sensor data.
- **RF Power Sensor**: Analog Devices AD8313 module, operating in the 100 MHz to 2.5 GHz range, with a dynamic range of up to 70 dB.
- **Magnetic Position Sensors**: Hall-effect magnetic sensors (KY-035 49E), used for homing reference detection on each rotational axis.
- **Power Supply**: Industrial 12 V DC, 2 A power supply (Orno OR-ZL-1631), rated at 24 W with IP20 protection and 230 V AC input, providing regulated voltage to all system components.
- **Mechanical Structure**: The platform consists of a one-meter fiberglass tube (non-conductive and lightweight), 3D-printed pulleys and brackets (PLA), and a laser-cut wooden base. Non-metallic materials were deliberately selected near the antenna to avoid reflections or distortions in the radiation pattern.

3.1.3 Electrical Subsystem

The electrical control system is based on an Arduino Uno board (ATmega328P), combined with a dual stepper motor driver shield. This shield integrates two DRV8825 drivers, each configured for 1/16 microstepping via onboard DIP switches. It simplifies the electrical connections and consolidates the power and control interfaces for the stepper motors.

The shield receives 12 V from a switched-mode power supply rated at 2 A. From this input, it distributes power to the stepper motors and also supplies the Arduino board.

Additionally, it powers auxiliary components, including the Hall-effect sensors and the RF power detector module. The onboard voltage regulation supports both 5 V and 12 V levels as required by each module.

Each of the two rotational axes includes a Hall-effect sensor positioned to detect a small magnet mounted on the corresponding gear. These sensors are used to perform an automated homing procedure during system initialization. A dedicated routine implemented in the firmware allows the system to precisely identify the reference position (zero-degree point) for both axes. This ensures reliable absolute positioning throughout the measurement process, compensating for the lack of closed-loop feedback in typical stepper motor systems.

The system wiring is clean and modular, facilitated by screwless terminals on the shield and the clear allocation of digital I/O pins (D4 to D8 and D12) for motor control. The shield also offers expansion capabilities, including I2C headers and XBee-compatible sockets, although no wireless features were used in the present implementation.

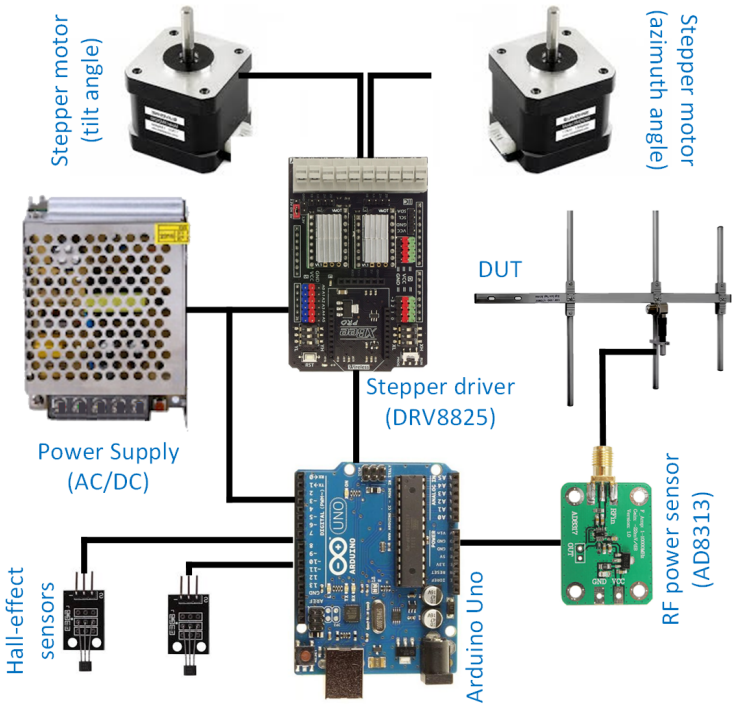


Figure 3.2: Block diagram of the electrical system, showing the Arduino Uno, dual DRV8825 driver shield, power supply, Hall-effect sensors, RF detector (AD8313), DUT and stepper motors.

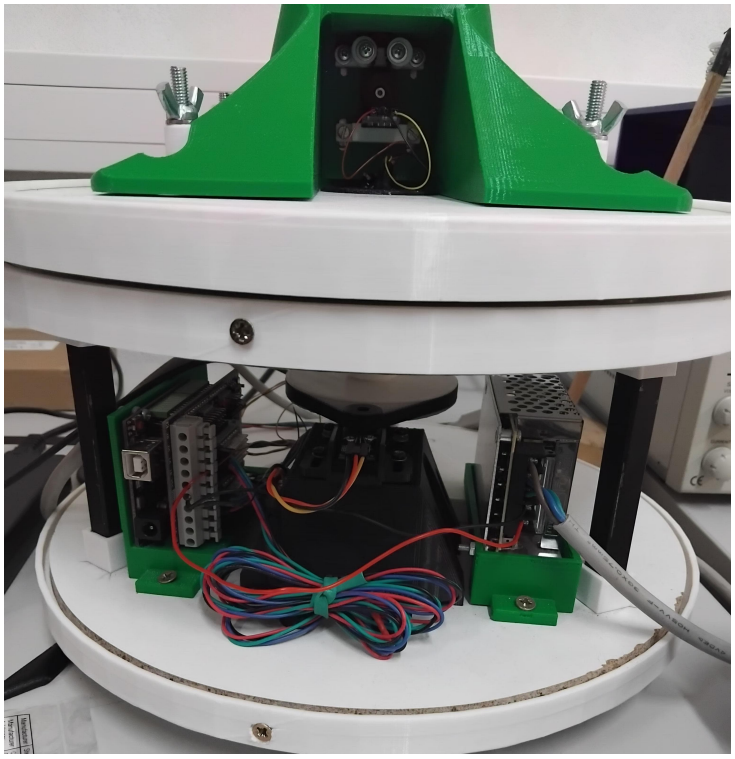


Figure 3.3: Assembled base of the measurement system showing the power supply, stepper driver shield, Arduino Uno, and internal wiring.

3.2 Hardware Configuration

The system assembly followed the schematic shown in Figure 3.4. The motors were connected to the Arduino Mega through the DRV8825 driver, configured for microstepping. The power sensor was integrated into the A13 analog port.

3.3 Software Development

The software was developed in two main parts:

- **Motor Control**: Implemented on the Arduino in C++ for precise axis movement.
- **Data Acquisition**: MATLAB based on the collecting and visualization of data from the sensor.



Figure 3.4: Electrical schematic of the measurement system.

3.4 Validation Strategy

To validate the system, experiments with a transmitting and receiving antenna positioned at different angles and distances were planned. Anechoic panels could be used to minimize reflections, and the results were compared with simulations performed in EMS software.

Chapter 4

Results

4.1 Introdução

Blá... blá...

Chapter 5

Conclusion and Future Work

Blá... blá...

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Appendix A

Título do Anexo

Listagem A.1: Exemplo de código.

```
1 setup(){
2 Serial.begin(115200);
3 }
4 loop(){
5 Serial.println("Ola");
6 delay(100);
7
8 }
```